Observability and Traffic Engineering

Istio Service Mesh on Bookinfo

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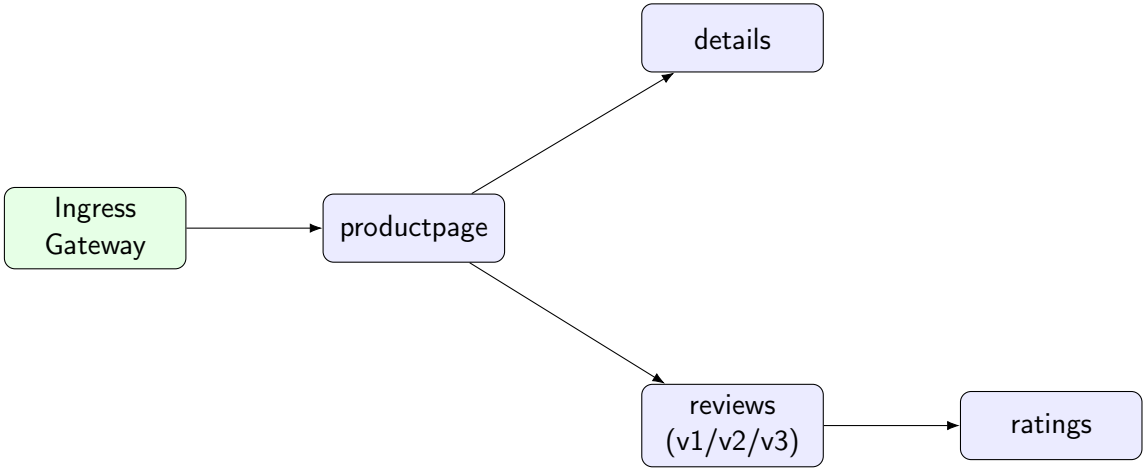
Agenda

1. Observability Stack
2. Traffic Engineering
3. Live Demonstration

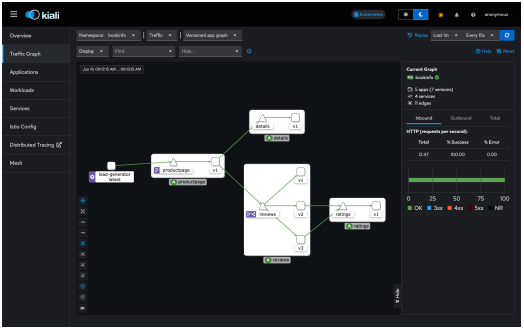
Mesh Architecture Under Observation



Every hop is an Envoy sidecar call: Prometheus scrapes metrics, Jaeger collects spans, and Kiali renders the live graph from both.

Kiali: Service Topology

- Renders the live traffic graph from Prometheus data, refreshed every 15s.
- Shows per-edge request rate and success %.
- Visualizes weighted routing (canary split) and header-based edges directly on the graph.
- **Istio Config** view lists every CRD (VirtualService, DestinationRule, Gateway, PeerAuthentication, Telemetry) with validation status.

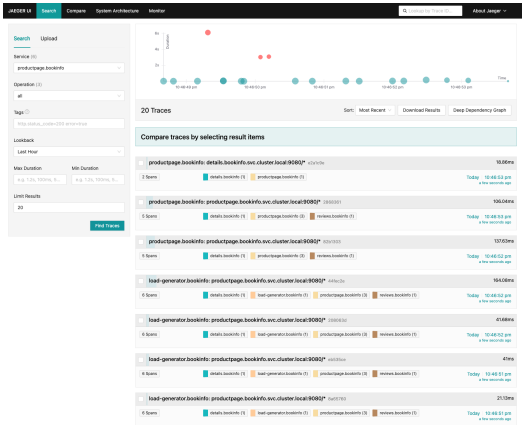


Grafana: Metrics Dashboards

- `06-install-metrics.sh` installs Prometheus and a custom **Bookinfo** dashboard.
- Three panels: request rate per service, 5xx error rate, and per-version `reviews` request rate.
- Standard Istio addon dashboards (Mesh, Service, Control Plane) also available.
- Used throughout the project to evidence canary splits and fault-injection scenarios.

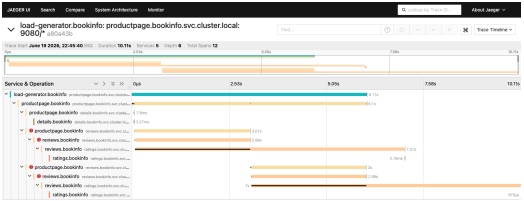
Jaeger: Distributed Tracing

- `07-install-tracing.sh` installs single-binary Jaeger plus a `Telemetry` resource setting **100% trace sampling** in `bookinfo` (vs. Istio's 1% default).
- Services propagate W3C/B3 headers end-to-end, so one trace spans every hop.
- Search view: every request entering at `productpage`, fanning out to `details` and `reviews`.



Jaeger: Trace Waterfall

- Full call tree: `load-generator` → `productpage` → {`details`, `reviews` → `ratings`}.
- 5 services, 12 spans, per-span timing.
- Pinpoints exactly which downstream call is slow or failing — e.g. a delayed `reviews` → `ratings` span surfaces immediately as an outlier.



Traffic Engineering: Overview

- Goal: control how requests to `reviews` are split across `v1`/`v2`/`v3` without touching application code.
- Two complementary mechanisms, both expressed as Istio `VirtualService` resources:
  1. **Canary release** — weighted traffic split.
  2. **Header-based routing** — deterministic routing for a specific user.
- Combined into a single `VirtualService` with priority-ordered HTTP match rules.

Canary Release: 90/5/5 Split

`networking/virtual-service-canary.yaml`

```
apiVersion: networking.istio.io/v1
kind: VirtualService
metadata:
  name: reviews-canary
spec:
  hosts: [reviews]
  http:
  - route:
    - destination: {host: reviews, subset: v1}
      weight: 90
    - destination: {host: reviews, subset: v2}
      weight: 5
    - destination: {host: reviews, subset: v3}
      weight: 5
```

`scripts/10-verify-canary.sh` samples 100 requests and asserts the v2+v3 ratio falls within $10\% \pm 5\%$.

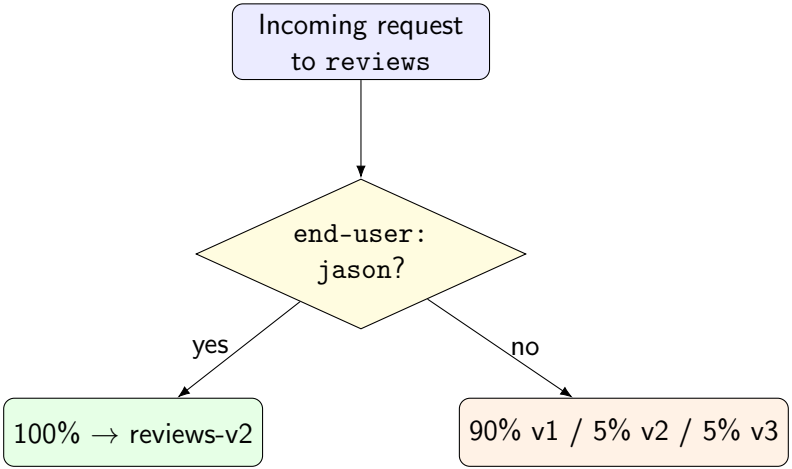
Header-Based Routing

`networking/combined-virtual-service.yaml` — priority order:

1. Header `end-user: jason` → 100% to `reviews-v2`.
2. Everything else → falls through to the 90/5/5 canary split.

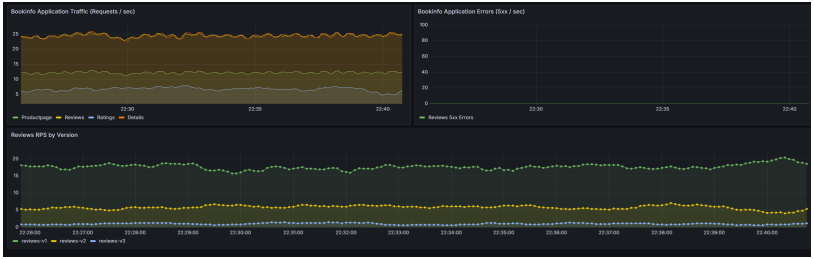
```
http:
  - name: header-based-routing
    match:
    - headers: {end-user: {exact: "jason"}}
    route:
    - destination: {host: reviews, subset: v2}
      weight: 100
  - name: canary-default-routing
    route:
    - destination: {host: reviews, subset: v1}
      weight: 90
    - destination: {host: reviews, subset: v2}
      weight: 5
    - destination: {host: reviews, subset: v3}
      weight: 5
```

Routing Decision Path



`scripts/11-test-header-routing.sh` confirms both paths: an anonymous request gets any valid version, a `jason` request always gets v2.

Observed Effect: Kiali Traffic Split



Kiali renders both the weighted edges to v1/v2/v3 and the header-matched edge to v2 under load.

Live Demonstration

1. Cluster, Istio, Bookinfo, metrics, and tracing already running (`scripts/run-all.sh`, lines 1–37).
2. Apply the **canary** `VirtualService` and run `10-verify-canary.sh`.
3. Apply the **combined header-based** `VirtualService` and run `11-test-header-routing.sh`.
4. Observe the effect live in **Kiali**, **Grafana**, and **Jaeger**.

No further slides — switching to the terminal and browser.

Thank You

Questions?